TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	90.5 ± 1.28
Correlation Stability	94.4 ± 0.49
Distribution Consistency	91.4 ± 1.20
Entropy Preservation	81.6 ± 2.69
Mean Z Score Drift	0.0 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	20.5 ± 1.12
Data Density	100.0 ± 0.00
Coverage Score	66.0 ± 2.68
Feature Accuracy	82.3 ± 3.29
Novelty Score	0.7 ± 0.01

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Dataset Audit: thoracic_surgery.csv

1. Dataset Portrait

Attribute	Value
Rows	470
Columns	17
Numeric features	17
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
DGN	3.096	3.074	-0.021	0.7%	OK
PRE4	3.282	3.252	-0.030	0.9%	OK
PRE5	4.569	6.317	+1.748	38.3%	Drift
PRE6	0.781	0.827	+0.046	5.9%	OK
PRE7	0.066	0.059	-0.007	11.3%	Warn
PRE8	0.145	0.160	+0.015	10.3%	Warn
PRE9	0.066	0.074	+0.009	12.9%	Warn
PRE10	0.687	0.715	+0.028	4.1%	OK
PRE11	0.166	0.162	-0.004	2.2%	OK
PRE14	11.736	11.737	+0.001	0.0%	OK
PRE17	0.074	0.074	+0.000	0.0%	OK
PRE19	0.004	0.000	-0.004	100.0%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	90.5 ± 1.28
Correlation Stability (*2)	94.4 ± 0.49
Distribution Consistency (*3)	91.4 ± 1.20
Entropy Preservation (*4)	81.6 ± 2.69
Mean Z Score Drift (*5)	0.0 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	20.5 ± 1.12

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	66.0 ± 2.68
Feature Accuracy (*10)	82.3 ± 3.29
Novelty Score (*11)	0.7 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	0.8553191489361703	0.8553191489361703	0	
F1	N/A	0.0	N/A	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.851063829787234	0.8510638297872342	0.85	-0.00106382978723406	
F1	0.037675637056442016	0.028717948717948714	0.02761904761904762	-0.010056589437394393	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8542553191489363	0.852127659574468	0.851063829787234	-0.003191489361702126	
F1	N/A	0.0	0.026666666666666665	0.02666666666666665	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	0.8553191489361703	0.8563829787234043	0.00106382978723404097	
F1	N/A	0.0	0.014285714285714285	0.01428571428571428 5	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8500000000000002	0.8414893617021277	0.8521276595744682	0.00212765957446807 7	
F1	N/A	0.01	N/A	0	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.808510638297872				

4

0.787234042553191

5

0.814893617021276

6

0.006382978723404255

3

F1	0.11354473527848138	0.14044054470827927	0.14791473083235096	0.03436999555386955	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8382978723404255	0.8202127659574467	0.8404255319148936	0.00212765957446809 95	
F1	0.08705128205128206	0.06327485380116958	0.11224373807494041	0.02519245602365807	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.21063829787234045	0.32234042553191494	0.2	-0.01063829787234047	
F1	0.25685562066118 9	0.23723526774481844	0.25003644841856897	-0.006819172242619312	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.847872340425532	0.852127659574468	0.847872340425532	-1.1102230246251566e-17	
F1	N/A	0.0	N/A	0	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8138297872340425	0.8010638297872345	0.8191489361702129	0.005319148936170215	
F1	0.11972345964178996	0.07079086605402395	0.15027777777777778	0.030554318135987822	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	2.88	2.16	1.0	0.0	0.0	0.0	1.0	1.0	14.0
2	3.0	3.4	1.88	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.76	2.08	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	3.68	3.04	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	2.44	0.96	2.0	0.0	1.0	0.0	1.0	1.0	11.0
6	3.0	2.48	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	4.36	3.28	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	2.0	3.19	2.5	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	3.16	2.64	2.0	0.0	0.0	0.0	1.0	1.0	11.0
10	3.0	2.32	2.16	1.0	0.0	0.0	0.0	1.0	0.0	11.0
11	3.0	2.56	2.32	0.0	0.0	1.0	0.0	1.0	0.0	12.0
12	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
13	3.0	3.0	2.36	1.0	0.0	0.0	0.0	1.0	1.0	11.0
14	2.0	3.98	3.06	2.0	0.0	0.0	0.0	1.0	1.0	14.0
15	3.0	1.96	1.4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
16	3.0	4.68	4.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
17	2.0	2.21	1.88	0.0	0.0	1.0	0.0	0.0	0.0	12.0
18	2.0	2.96	1.67	0.0	0.0	0.0	0.0	0.0	0.0	12.0
19	3.0	2.6	1.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
20	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	5.0	2.50478811526738 2	73.2324389929518 6	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	4.0	4.88019982212819 4	67.8731649567457	1.0	0.0	0.0	0.0	0.0	0.0	11.0
3	3.0	2.10231741079884 6	2.61574007645270 26	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	4.48940638873245 1	16.4470867629416 4	1.0	0.0	0.0	0.0	0.0	0.0	12.0
5	3.0	3.53144274337470 5	2.5104950101903	1.0	0.0	0.0	0.0	1.0	0.0	14.0
6	3.0	3.94590572925165 6	9.36055657165716	0.0	0.0	0.0	0.0	0.0	0.0	12.0
7	3.0	2.68	2.16	0.0	0.0	0.0	0.0	0.0	0.0	12.0
8	4.0	2.32	1.76	1.0	0.0	1.0	0.0	1.0	1.0	11.0
9	2.0	2.72	2.08	0.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	4.23413600360406 5	2.17270059927375 45	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	4.0	4.14245126941025 9	11.5960373454593 81	1.0	1.0	0.0	0.0	1.0	0.0	11.0
3	3.0	2.60739607235741 97	2.44712249160056 2	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	4.0	3.05804848380621 94	2.85041599082742 7	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	2.87558609141196 26	2.41552831363801 12	1.0	0.0	0.0	1.0	1.0	0.0	11.0
6	3.0	1.82351940780934 75	23.8397042819292 8	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	2.84	2.24	1.0	1.0	1.0	0.0	0.0	0.0	12.0
8	3.0	3.36	2.8	1.0	0.0	0.0	0.0	1.0	1.0	12.0
9	3.0	2.44	1.96	1.0	0.0	1.0	1.0	1.0	0.0	13.0
10	3.0	5.16	4.28	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.34056664692505 5	2.12182357072685 02	1.0	0.0	0.0	0.0	1.0	1.0	12.0
2	3.0	3.64568137058898 5	2.72838306441775 02	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	3.35506105745831 24	3.06106454822719 03	0.0	0.0	0.0	0.0	0.0	0.0	11.0
4	3.0	3.11356435725389	1.75254799684908 18	0.0	0.0	0.0	0.0	1.0	0.0	11.0
5	3.0	3.08490496010937 54	2.11216956217971 94	1.0	0.0	0.0	0.0	1.0	1.0	11.0
6	3.0	3.37819225059641	3.13838675321403 53	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	4.0	4.24	3.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	3.87	2.68	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	5.0	3.67	76.8	0.0	1.0	1.0	0.0	1.0	0.0	12.0
10	3.0	3.8	2.84	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	4.0	3.10242840446760 6	2.04008138958907 2	1.0	0.0	0.0	0.0	1.0	0.0	11.0
2	3.0	0.96224025169151 41	1.70253196103655 48	1.0	0.0	0.0	0.0	1.0	1.0	12.0
3	3.0	4.21691279172895 8	2.24920496825283 32	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	4.0	3.61260914966112 8	2.11693734084359 74	1.0	1.0	0.0	0.0	0.0	0.0	12.0
5	3.0	3.68570046388925 74	2.07664720469505 46	1.0	0.0	0.0	0.0	0.0	0.0	12.0
6	4.0	1.70975202163226 04	1.12743546056309 2	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	1.84	1.28	1.0	0.0	0.0	0.0	1.0	1.0	11.0
8	5.0	4.95	4.12	1.0	0.0	0.0	0.0	0.0	1.0	11.0
9	3.0	2.92	2.28	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	3.67002639305325 1	3.13595180725785 3	1.0	0.0	0.0	0.0	1.0	1.0	13.0
2	4.0	3.88832240966758 5	3.00865682146039 8	2.0	0.0	0.0	0.0	1.0	1.0	11.0
3	3.0	2.40787600659298 9	4.61772951324027 4	1.0	0.0	0.0	0.0	0.0	0.0	11.0
4	3.0	2.18948336543681 47	1.73646665530493 26	0.0	0.0	0.0	0.0	0.0	0.0	12.0
5	2.0	2.77400840925668 1	1.71174802382909 56	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	2.66688480484618 4	2.85028884994556 5	0.0	0.0	0.0	0.0	0.0	0.0	11.0
7	3.0	3.0	2.44	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	2.4	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	4.36	3.92	1.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	3.4	2.8	1.0	1.0	0.0	0.0	1.0	1.0	11.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	1.82540129127209 72	1.22176484939897 88	1.0	0.0	1.0	0.0	0.0	1.0	11.0
2	4.0	4.78732955524399 5	3.22534541111857 86	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.52365780412112 17	2.54850472374497 4	0.0	0.0	0.0	0.0	1.0	1.0	11.0
4	3.0	3.15777044771733 8	1.84775398894145 14	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	5.0	3.28679137353603 9	2.58785940399582 8	0.0	0.0	0.0	0.0	0.0	0.0	11.0
6	3.0	2.50901343580168 2	1.64898000708956 48	2.0	0.0	0.0	1.0	1.0	1.0	12.0
7	2.0	3.24	2.52	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.4	1.64	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	3.24	52.3	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	4.0	4.78913902474940 2	2.86888587678018 6	0.0	0.0	0.0	0.0	1.0	0.0	12.0
2	4.0	5.25841749698233 8	21.8413898038768 44	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.40110981842706 4	2.09355296750537 43	0.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	4.99492674911054 5	36.5999059607301 26	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	2.86196934256423	2.25105640784273 84	0.0	0.0	0.0	0.0	1.0	0.0	11.0
6	3.0	3.61790905782015 5	2.39806083362475 1	2.0	0.0	1.0	0.0	1.0	1.0	11.0
7	3.0	2.84	2.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	2.2	1.44	1.0	0.0	0.0	0.0	1.0	0.0	12.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	6.00513781038999 7	51.6595436239563 9	0.0	0.0	0.0	0.0	0.0	0.0	12.0
2	3.0	1.72611647794246 65	0.31482104693246 88	1.0	0.0	1.0	0.0	1.0	1.0	11.0
3	3.0	4.59306831959743 9	3.41435878871076 87	1.0	0.0	0.0	1.0	1.0	0.0	12.0
4	3.0	4.24746816612902 7	3.53632149649612 56	1.0	0.0	0.0	1.0	0.0	1.0	12.0
5	3.0	2.82984894466170 3	1.42191371575997 15	0.0	0.0	0.0	0.0	1.0	0.0	11.0
6	3.0	2.45103097251712 44	1.83516836232262 14	1.0	0.0	0.0	0.0	1.0	1.0	11.0
7	3.0	2.76	2.28	0.0	0.0	0.0	0.0	0.0	0.0	11.0
8	4.0	5.03	79.3	1.0	0.0	0.0	1.0	0.0	0.0	11.0
9	2.0	2.16	1.56	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	11.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	3.5568906269034306	2.050872277151387	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	3.17014752371294	4.6502789404052445	1.0	0.0	0.0	1.0	1.0	0.0	11.0
3	5.0	1.3894387887728676	62.737826560802596	1.0	0.0	0.0	0.0	0.0	0.0	11.0
4	3.0	2.363809061981835	1.7429196449447373	1.0	1.0	0.0	0.0	0.0	0.0	12.0
5	3.0	2.1757217026527775	1.3383950950720167	1.0	0.0	0.0	0.0	1.0	1.0	12.0
6	3.0	3.7835127553945798	1.9628716810787936	1.0	0.0	0.0	0.0	1.0	1.0	12.0
7	3.0	3.44	3.13	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	3.64	3.12	1.0	0.0	0.0	0.0	1.0	0.0	12.0
9	3.0	2.56	60.9	0.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	2.8	3.36	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	4.0	2.6293111827926396	3.6575501357684272	1.0	0.0	0.0	0.0	0.0	0.0	11.0
2	3.0	2.7935366307978313	2.5350660669906917	1.0	1.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.3885591413134209	1.8288591479703027	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	2.5869746718416997	1.7757576158376367	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	3.0768706294529125	-12.207996195930631	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	3.629018005636427	4.114762503006642	0.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	3.48	2.52	1.0	0.0	0.0	0.0	1.0	0.0	14.0
8	3.0	4.48	3.48	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	3.0	2.63	67.3	1.0	0.0	0.0	1.0	1.0	0.0	11.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0